CUSTOMER PURCHASE PREDICTION

USING REGRESSION AND RANDOM FOREST

Machine Learning Project

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B.Tech | A.Y. 2026–27

2. Introduction

- Customer Purchase Prediction uses Machine Learning to predict whether a customer is likely to purchase a product or service.
- Customer information such as age, gender, income, previous purchases, website visits and browsing activity can reveal purchasing patterns.
- Historical customer data can help businesses improve marketing strategies and business decisions.
- Project focus: learn patterns from customer data and generate predictions for new customers.

3. Problem Statement

- Businesses have large amounts of customer information but may find it difficult to identify potential buyers.
- Manual analysis is time-consuming and can miss useful patterns in customer behavior.
- Using the same marketing strategy for every customer may lead to unnecessary marketing effort.
- Need: a Machine Learning system that can analyze customer attributes and predict purchasing behavior.

4. Motivation

- Customer behavior is important for business growth and marketing decisions.
- Machine Learning can identify patterns in historical customer data.
- Prediction can support targeted advertising and personalized offers.
- The project provides practical experience with the complete Machine Learning workflow.

5. Objectives

- Main objective: predict customer purchasing behavior using Regression and Random Forest.
- Collect and understand customer purchase data.
- Preprocess and clean the dataset.
- Select relevant customer features.
- Train and evaluate Regression and Random Forest models.
- Use the trained models to predict purchase behavior for new customers.

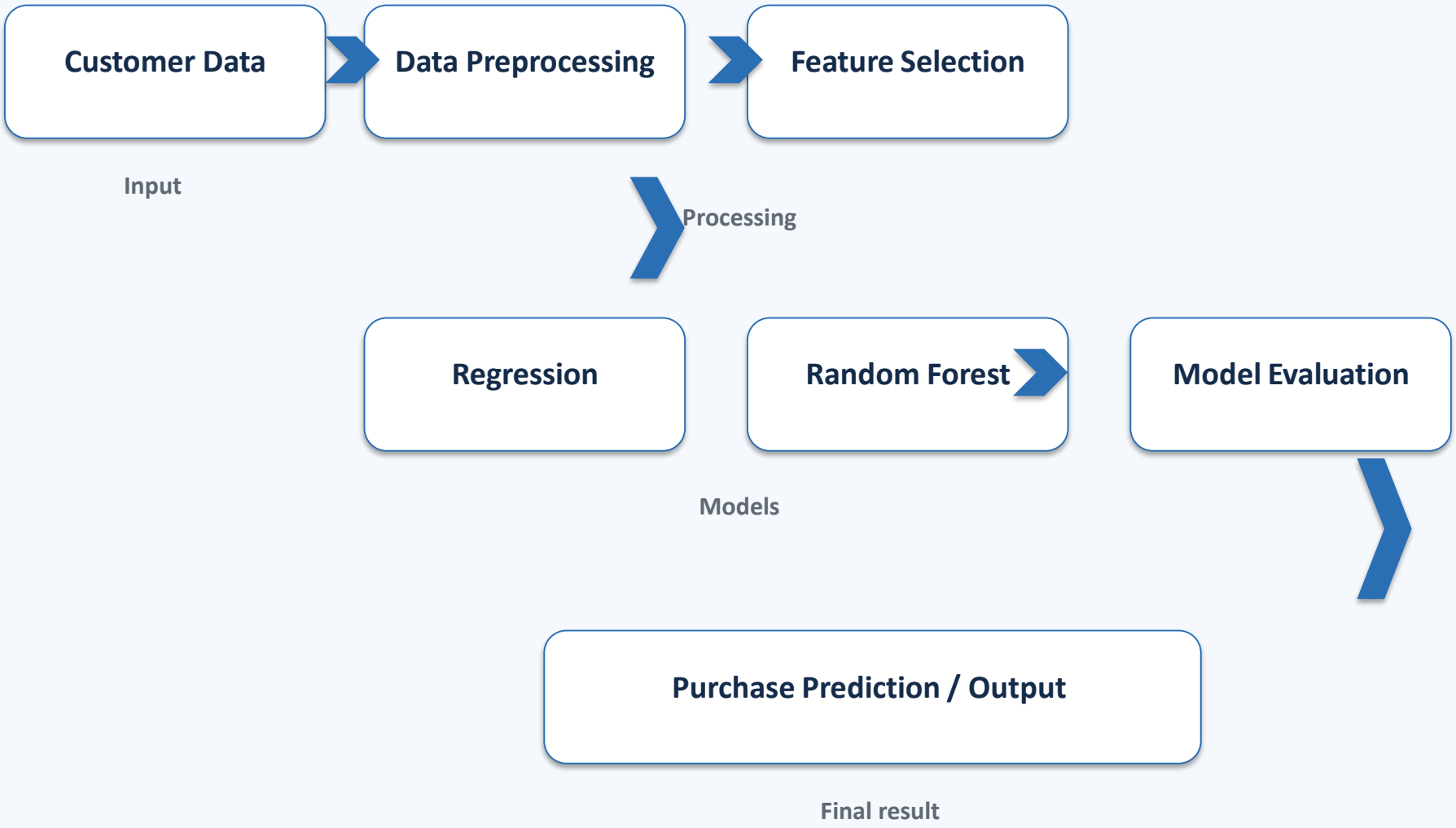
6. Existing System

- Traditional approaches may depend on manual analysis, basic customer segmentation or general marketing campaigns.
- These approaches can be difficult to scale when customer data becomes large.
- Customer-specific purchasing patterns may not be identified efficiently.
- Marketing resources may be spent on customers who are less likely to purchase.

7. Proposed System

- Build a Machine Learning system using historical customer information.
- Clean and preprocess the customer dataset.
- Use relevant features to train Regression and Random Forest models.
- Test the models on unseen customer records.
- Compare model performance using suitable evaluation metrics.
- Generate predictions that can support targeted marketing and business decisions.

8. System Architecture



9. Methodology / Workflow

1. Data Collection – obtain the customer purchase dataset.
2. Data Preprocessing – handle missing values, duplicates and categorical values.
3. Feature Selection – retain relevant customer attributes.
4. Train/Test Split – divide data into training and testing sets.
5. Model Training – train Regression and Random Forest.
6. Evaluation – test models using unseen customer data and suitable metrics.
7. Prediction – generate purchase predictions for new customers.

10. Technology Stack

- Operating System: Windows 10/11
- Programming Language: Python 3.x
- Development Tool: Jupyter Notebook / VS Code
- Data Processing: Pandas, NumPy
- Machine Learning: Scikit-learn
- Data Visualization: Matplotlib, Seaborn
- Dataset: Customer Purchase Dataset
- Models: Regression and Random Forest

11. System Modules / Components

- Data Collection Module – obtains customer information.
- Data Preprocessing Module – cleans and transforms the data.
- Feature Selection Module – identifies relevant customer attributes.
- Machine Learning Module – trains Regression and Random Forest models.
- Prediction Module – predicts customer purchase behavior.
- Evaluation Module – measures and compares model performance.
- Visualization Module – presents graphs and results.

12. Implementation

- Import Python libraries required for data processing, modeling and visualization.
- Load the customer dataset and inspect its structure.
- Clean the dataset and handle missing values.
- Convert categorical features into numerical representations.
- Select relevant features and split the data into training and testing sets.
- Train Regression and Random Forest models.
- Generate predictions and evaluate the trained models.
- Display results using suitable tables and graphs.

13. Dataset / Input Data

- Dataset: Customer Purchase Dataset.
- Project data includes customer attributes such as age, gender, income, previous purchases, website visits and browsing activity.
- Target: customer purchase behavior.
- Preprocessing includes missing-value handling, duplicate removal when required, categorical encoding and feature selection.
- Important: the abstract does not specify the exact number of records or exact feature count; these should be added from the actual dataset.

14. Algorithm / Model

Regression

Identifies relationships between input variables and a target variable. Linear Regression can be used for continuous targets; Logistic Regression can be used for binary purchase / no-purchase outcomes.

Random Forest

An ensemble method that combines multiple decision trees. It can handle multiple features and complex or nonlinear relationships.

Why compare them?

The project demonstrates two different modeling approaches and evaluates their performance on unseen customer data.

Model → Prediction → Evaluation → Comparison

15. Results

- ✓ Both trained models are tested using unseen customer records.
- ✓ Suitable evaluation metrics can include Accuracy, Precision, Recall and F1-score when the target is a classification outcome.
- ✓ Model performance can be compared using the actual results obtained from the implementation.
- ✓ Do not report an accuracy or score unless it has been obtained from the project implementation.

16. Screenshots / Demo

Recommended screenshot 1: Customer dataset.

Recommended screenshot 2: Data preprocessing / cleaned dataset.

Recommended screenshot 3: Model training in Jupyter Notebook or VS Code.

Recommended screenshot 4: Prediction output for a customer.

Recommended screenshot 5: Graph comparing model performance.

17. Advantages

- Predicts customer purchasing behavior from historical data.
- Helps identify potential customers.
- Supports targeted marketing and personalized offers.
- Can reduce unnecessary marketing effort.
- Uses multiple customer attributes for prediction.
- Supports data-based business decisions.
- Demonstrates a complete practical Machine Learning workflow.

18. Applications

- E-commerce – identify customers likely to purchase products.
- Digital marketing – support targeted advertising.
- Retail – understand customer purchasing patterns.
- Customer Relationship Management – support personalized offers.
- Sales planning – use predictions to support business decisions.
- Target users include e-commerce businesses, retailers, marketing teams and sales teams.

19. Limitations

- ✓ Prediction quality depends on the quality of the dataset.
- ✓ Historical data may not always represent future customer behavior.
- ✓ Missing or incorrect data can affect model predictions.
- ✓ Customer behavior can change over time.
- ✓ Model performance depends on the selected features and algorithms.
- ✓ The current project is based on the available customer dataset.

20. Future Scope

- Use larger and more diverse customer datasets.
- Add more customer behavior features.
- Explore additional Machine Learning algorithms.
- Improve feature engineering and model performance.
- Develop a web-based customer purchase prediction system.
- Deploy the trained model as an online service.
- Support real-time customer predictions.

21. Conclusion

- The project demonstrates how Machine Learning can be applied to customer purchase prediction.
- Historical customer information is preprocessed and used to train models.
- Regression provides an interpretable modeling approach.
- Random Forest provides a flexible approach for multiple features and complex relationships.
- The system can support customer targeting, personalized offers, marketing decisions and sales planning.
- Overall, the project provides practical experience in data preprocessing, model training, prediction and evaluation.